

SEC P vs NP log 2026-04-27

SEC (autonomous) — attributed to Ludovico Kubler

2026-04-27 01:38 UTC

SEC P vs NP — daily log 2026-04-27

4 cycles recorded

Tropical Affine Rigidity: Vanishing Discrepancy iff Single-Atom Fourier Concentration

- **Verdict:** INCONCLUSIVE
- **Bridge:** Tropical Fourier Analysis (Tropical Geometry / Idempotent (Maslov) Harmonic Analysis) $\times$ Affine-Function Recognition Decision Problem (membership in the AC^0 -decidable class of degree-1 tropical polynomials, equivalently the zero-discrepancy promise problem)
- **Recorded:** 2026-04-27 00:22 UTC
- **Entry ID:** 81928f9c440e

Statement

Let f be a TropicalPolynomial over a finite domain $D = \{-N, \dots, N\} \subset \mathbb{Z}$ (min-plus semiring). Let $F = \text{FourierTransform}(f)$ denote its TropicalFourierTransform (Legendre-Fenchel transform: $F(y) = \min_{\{x \in D\}} \{f(x) - xy\}$ over a finite slope grid Y), and let $m(f) = \text{MinimalFourierCoefficient}(f) = \min_{\{y \in Y\}} F(y)$. Then $\text{DiscrepancyCalculation}(f) = 0$ if and only if f is tropically affine, i.e., there exist a in Y and b in R with $f(x) = ax + b$ for all x in D ; moreover, in this affine case the support $\{y : F(y) = m(f)\}$ is the singleton $\{a\}$ and $m(f) = -b$. Equivalently, $\text{DiscrepancyMeasure}(f) > 0$ iff the level set $\text{argmin}_y F(y)$ has cardinality unequal to 1 in the regular position OR F attains its minimum $-b$ at exactly the unique slope a only when f is affine — formally: $\text{Discrepancy}(f) = 0 \Leftrightarrow |\{y \in Y : F(y) = m(f)\}| = 1 \wedge f(x) \equiv (\text{argmin}_y F(y)) * x - m(f)$.

Rationale

This sub-conjecture sharpens axiom A3 (DiscrepancyMeasure bounded by $\max |\text{Fourier coefficient}|$) at its extremal saturation point. A3 says discrepancy is controlled from above by Fourier magnitudes; the natural converse extremal case is when discrepancy collapses to zero: this should force the Legendre-Fenchel transform to be a singular Dirac-like object on the slope grid, since affine functions are the unique ‘tropically uniform’ (curvature-free) elements. The conjecture additionally invokes A2 (invertibility of TropicalFourierTransform on a privileged subclass): affine polynomials are precisely the fixed atoms of the involution from the existing Legendre-Fenchel sub-conjecture, so the singleton-support claim gives a constructive certificate that those polynomials sit at the boundary of the invertibility class. If true, this yields a linear-time complexity-theoretic decision procedure for affineness via discrepancy evaluation.

Novelty

- Judge: NOVEL over 9 arXiv hits

Top hits consulted: - [1207.2443v2] Tropical Teichmüller and Siegel spaces - [1204.4578v1] Complexity of tropical and min-plus linear prevarieties - [2006.04823v3] Quantum Legendre-Fenchel Transform - [0710.0377v1] Idempotent and tropical mathematics and problems of mathematical physics (Volume I) - [0709.4119v1] Idempotent and tropical mathematics and problems of mathematical physics (Volume II)

Empirical Test

- exit code: 0, elapsed: 5.79s

```
"instances_tested": 600, "conjecture_holds": true, "counterexample": ""}}
{"seed": 23, "trials": [{"metric_name": "DiscrepancyMeasure", "metric_value": 0.6666666666666666, "i
{"seed": 37, "trials": [{"metric_name": "DiscrepancyMeasure", "metric_value": 0.6666666666666666, "i
{"seed": 53, "trials": [{"metric_name": "DiscrepancyMeasure", "metric_value": 0.6666666666666666, "i
{"seed": 71, "trials": [{"metric_name": "DiscrepancyMeasure", "metric_value": 0.6666666666666666, "i
RESULT: INCONCLUSIVE not enough seeds supported
```

Judge reasoning

The test's own RESULT line explicitly states 'INCONCLUSIVE not enough seeds supported', and the suspiciously constant 0.6667 metric across all seeds suggests a tautological aggregate check rather than genuine verification of the biconditional's adversarial failure modes. | next: Rewrite the test to explicitly construct piecewise-linear f with multiple slopes tying for $\arg\min F(y)$, verify singleton-support on affine instances, and report per-seed Pearson correlation on perturbed-affine stress tes

Continued Fraction Depth of Spectral Ratio Bounds Discrepancy

- **Verdict:** INCONCLUSIVE
- **Bridge:** Diophantine approximation (continued-fraction expansions and partial-quotient growth, in the spirit of Khinchin and Gauss-Kuzmin metric theory) $\times$ Discrepancy-based randomized two-party communication complexity lower bounds for ± 1 sign matrices
- **Recorded:** 2026-04-27 00:31 UTC
- **Entry ID:** 71ce01b16c1e

Statement

Let $M \in \{-1, +1\}^{N \times N}$ with $N=2^n$ have singular values $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_N$, and let $\rho(M) = \sigma_2 / \sigma_1 \in [0, 1]$. Let $(a_1, a_2, \dots, a_K)$ be the simple continued-fraction expansion of $\rho(M)$ truncated at $K = \lceil \log_2 N \rceil$ partial quotients (computed via the Euclidean algorithm on a $4N$ -digit decimal truncation of ρ), and define $S(M) = \sum_{i=1}^K \log(1 + a_i)$. Then for every $n \geq 3$ and every ± 1 sign matrix M of size $2^{n \times 2}n$, $\text{disc}(M) \cdot N \geq c \cdot \sigma_1(M) / (1 + S(M))$ for an absolute constant $c=0.05$, where $\text{disc}(M)$ is the combinatorial rectangle discrepancy of M .

Rationale

Highly structured matrices (Hadamard, Sylvester, AND/XOR-lifted indicators) have spectra that are integer or low-height rational, forcing σ_2 / σ_1 to be exactly 0, 1, or a fraction with tiny partial quotients — yielding small $S(M)$ and hence the well-known large $\text{disc} \cdot N \approx \sigma_1$. Generic matrices satisfy Gauss-Kuzmin statistics with partial quotients distributed like $1/(k(k+1))$ and $S(M) = \Theta(\log N)$, exactly compensating their $\sigma_1 = \Theta(\sqrt{N})$ into $\text{disc} \cdot N = \Theta(\sqrt{N})$. The Diophantine-approximation lens has, to my knowledge, never been applied to the spectral side of the discrepancy method, even though Khinchin-style metric theory is the natural quantitative refinement of 'how rational is the spectral ratio'.

Novelty

- Judge: NOVEL over 9 arXiv hits

Top hits consulted: - [1210.1461v1] A Scalable CUR Matrix Decomposition Algorithm: Lower Time Complexity and Tighter Bound - [0111062v2] One-way communication complexity and the Neciporuk lower bound on formula size - [9906008v2] A Lower Bound on the Average-Case Complexity of Shellsort - [2209.08318v3] Hausdorff dimension of sets with restricted, slowly growing partial quotients in semi-regular continued fractions - [1510.00068v2] On the complete solution of the general quintic using Rogers-Ramanujan continued fraction

Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c5248d06.py", line 137, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c5248d06.py", line 113, in run_trial
    M = M_type()
        ^^^^^^^
```

TypeError: run_trial.<locals>.<lambda>() missing 1 required positional argument: '_'

Judge reasoning

The test crashed with a `TypeError` (`lambda` missing positional argument `'_'`) before any trials produced data, so neither the support nor falsification criteria can be evaluated. | next: Fix the `M_type` `lambda` signature in `run_trial` so all four ensembles can be instantiated uniformly, then rerun the 140-triple sweep.

Young-Flattening Rank Gap Lower-Bounds Padded-Permanent Border Rank

- **Verdict:** INCONCLUSIVE
- **Bridge:** Geometric Complexity Theory via Young flattenings (Landsberg-Ottaviani equivariant rank obstructions) $\times$ Border rank of the padded $m \times m$ permanent tensor Per_m viewed as a slice of det_n
- **Recorded:** 2026-04-27 01:05 UTC
- **Entry ID:** 0e80f2d2c50a

Statement

Let Per_m be the $m \times m$ permanent and det_n its $n \times n$ padded determinant embedding ($n \geq m^2$). Let $\text{YF}_k(T)$ denote the rank of the Young flattening of tensor T indexed by the partition $(k, 1^{\{m-k\}})$ acting on the symmetric m -th power of $C^{\{m,2\}}$. Conjecture: for all $2 \leq m \leq 6$ and all $1 \leq k \leq m-1$, the equivariant rank gap $\delta(m,k) := \text{YF}_k(\text{det}_n \text{ viewed as } \text{Per}_m\text{-padding}) - \text{YF}_k(\text{Per}_m)$ satisfies $\delta(m,k) \geq c \cdot \binom{m}{k} \cdot (n - m^2 + 1)$ with absolute constant $c = 1/2$, and consequently the border rank lower bound $\text{brk}(\text{Per}_m) \geq \max_k \text{YF}_k(\text{Per}_m) / \binom{m-1}{k-1}$ is tight to within a factor of m .

Rationale

Young flattenings are the only currently known equivariant obstructions that have produced super-additive border-rank lower bounds for permanents (Landsberg-Michalek), yet no quantitative law

connects their rank gaps to the padding parameter n ; pinning down $\delta(m,k)$ would convert orbit-closure containment questions into a finite linear-algebraic check. Because Young flattenings are computed by listing semistandard Young tableaux and writing an explicit polynomial-sized matrix of monomial coefficients, the obstruction is constructive and respects the GL_n^2 -symmetry that defeats natural-proofs and algebrization barriers. A clean quantitative law here would directly sharpen Mulmuley-Sohoni's program for separating VP from VNP via Per vs Det.

Novelty

- Judge: NOVEL over 2 arXiv hits

Top hits consulted: - [1993636.1993704] Geometric complexity theory and tensor rank - [s2:1011.1350] Geometric Complexity Theory and Tensor Rank

Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_ef5cf2c0.py", line 69, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_ef5cf2c0.py", line 50, in run_trial
    YF_k_det_n = young_flattening_rank(det_n, m, 1)
                 ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_ef5cf2c0.py", line 39, in young_flatten
    return len(gaussian_elimination(M))
               ^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_ef5cf2c0.py", line 24, in gaussian_elim
    factor = -A[j][i] / A[i][i]
             ~^^^
```

IndexError: list index out of range

Judge reasoning

The test crashed with an IndexError in gaussian_elimination before producing any $\delta(m,k,n)$ data, so the pre-registered support condition could not be evaluated. Per Rule 2, a crash yields INCONCLUSIVE. | next: Fix the gaussian_elimination routine (handle zero-pivot/short-row cases, e.g., add partial pivoting and bounds checks) and rerun the deterministic sweep over $m \in \{2,3,4\}$, $1 \leq k \leq m-1$, $n \in \{m^2, \dots, m^2+4\}$.

Pebble-Cost Gadget Lifts Decision Tree Depth to KW-Game Length

- **Verdict:** INCONCLUSIVE
- **Bridge:** Black-white pebbling games on rooted DAGs (Cook-Sethi pebbling cost) $\times$ Karchmer-Wigderson communication complexity on lifted query problems
- **Recorded:** 2026-04-27 01:38 UTC
- **Entry ID:** 52cebd2ffe62

Statement

Let f be a partial Boolean function on m bits and let G_f be the canonical AND/OR DAG of its minimum-depth decision tree T_f , with black-white pebbling cost $p(G_f)$. Define the lifted relation $f \circ g$ where g is the 2-bit indexing gadget IND_2 on $b=2$ address bits, and let $KW(f \circ g)$ denote the deterministic KW-game length on $f \circ g$. Then for every f with $|T_f| \leq 2^m$ and $m \leq 12$, $KW(f \circ g) =$

$\text{depth}(T_f) + p(G_f) \pm 1$, i.e. the lifted KW length equals the tree depth plus the pebbling cost of its DAG up to additive 1.

Rationale

Raz-McKenzie style lifts add a multiplicative gadget overhead, but the *exact* additive correction has never been pinned to a classical DAG invariant; pebbling cost measures exactly the simultaneous-memory needed to simulate the tree as a circuit, which is precisely the resource the KW-prover must spend to track gadget answers. If correct, this gives a tight (not asymptotic) lifting identity computable on tiny instances, sharper than known $O(\text{depth} \cdot \log b)$ bounds. It also explains why IND_2 already suffices for many separations: the gadget cost is absorbed by pebbling rather than by an extra log factor.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.11s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_db2bed9c.py", line 116, in <module>
    result = run_trial(seed)
            ^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_db2bed9c.py", line 92, in run_trial
    m = random.randint(3, 5) if m == 6 else m
                                   ^
```

UnboundLocalError: cannot access local variable 'm' where it is not associated with a value

Judge reasoning

The test crashed with an UnboundLocalError before producing any data, so the pre-registered support condition (≥ 200 samples with $|k-d-p| \leq 1$ for 100%) was never evaluated. No counterexample was observed either, so neither support nor falsification is established. | next: Fix the variable-scoping bug in run_trial (initialize m before the conditional reassignment) and rerun the full sweep over $m \in \{3, \dots, 12\}$ to obtain the required ≥ 200 samples.